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Course Welcome

Module 1 Introduction

Lesson 1: Introduction to deep learning

Lesson 2: Neural network

Lesson 3: Backpropagation

Module 1 Summative Assessment

Graded Assignment: Module 1 Summative Assessment Submitted

Module 1 Summary

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Module 1 Summative Assessment

Review Learning Objectives

Instructions

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

New session Let's chat

Assignment details

Due Nov 24, 11:59 PM IST

Attempts 0 left (1 attempts every 8 hours)

Retry

Submitted Nov 23, 2:24 AM IST

Your score: 100%

To pass you need at least 80%. We keep your highest score.

View submission See feedback

1. Which of the following best describes the role of a neuron in a neural network? 1 / 1 point

- A function that defines the network structure
- A data structure for storing training examples
- An entity that processes input and produces an output based on the input and its parameters
- A specific type of network architecture

Correct
Correct.

2. What does the chain rule enable in the context of backpropagation? 1 / 1 point

- Decreasing the computational load
- Calculation of gradients for parameter of each neuron
- Direct computation of the loss function Feedback:
- Elimination of the need for activation functions

Correct
Correct.

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3. Which application is NOT listed as a deep learning application in the slides? 1 / 1 point

- Face detection
- Smile detection
- AlphaGo
- Quantum computing

Correct
Correct.

4. Which of the following networks introduced the concept of "depth" in neural networks through many hidden layers? 1 / 1 point

- VGG
- GoogleNet
- ResNet
- All of the above

Correct
Correct.

5. What role does the softmax function play in a neural network? 1 / 1 point

- It acts as a regularization technique.
- It speeds up matrix operations.
- It serves as a multi-class classifier at the output layer.
- It initializes the weights of the neural network.

Correct
Correct.

6. Who won the Turing Award for the contribution to the deep neural networks 1 / 1 point

- Yoshua Bengio
- Geoffrey Hinton
- Yann LeCun
- All of above

Correct
Correct

7. What is the primary purpose of the backpropagation algorithm in neural networks? 1 / 1 point

- To increase the speed of the forward pass
- To optimize the network architecture
- To adjust the weights of the network based on the loss gradient
- To decrease the size of the input data

Correct
Correct

8. What is the role of gradient descent in the learning process of deep learning models? 1 / 1 point

- To calculate the exact loss for each example
- To randomly initialize model parameters
- To iteratively adjust model parameters to minimize loss
- To increase the complexity of the model

Correct
Correct.

9. What indicates the completion of the forward pass in backpropagation? 1 / 1 point

- When all parameters are initialized
- When the network architecture is defined
- When the outputs for all parameters are computed
- When the gradients for all parameters are calculated

Correct
Correct.

10. What is the primary purpose of backpropagation in neural networks? 1 / 1 point

- To backup weights of a neural network
- To increase the number of hidden layers in a neural network
- To efficiently compute the gradient of the loss function with respect to the weights
- To initialize the weights of a neural network randomly

Correct
Correct.

11. Which model below has the best performance on ImageNet competition? 1 / 1 point

- AlexNet
- VGG
- GoogleNet
- ResNet

Correct
Correct.

12. What is the primary programming language recommended for projects in this deep learning course? 1 / 1 point

- Java
- C++
- Python
- R

Correct
Correct!

13. What is the primary purpose of using a loss function in deep learning models? 1 / 1 point

- To increase the accuracy of the model on training data
- To measure the goodness of the model's predictions,
- To optimize the model's network structure
- To prevent the model from overfitting

Correct
Correct.

14. Which technique is mentioned as a way to speed up matrix operations in neural networks? 1 / 1 point

- Gradient descent
- Backpropagation
- Parallel computing
- Regularization

Correct
Correct.

15. In which year, deep learning become popular in image analysis? 1 / 1 point

- 2009
- 2011
- 2012
- 2015

Correct
Correct.